

Chapter 5

Vibration of String and Wave Propagation

Strings are naturally musical sound instruments as are acoustic pipes. This is because free vibration of one-dimensionally extending media such as a string can be composed of a fundamental and its harmonics, which are important for musical sound and with scales. Scientific studies on the resonance of string vibrations date from Pythagoras, and Galileo mentioned eigenfrequencies of free vibration of a string. This chapter starts by focussing on progressive waves and their propagation speed along a long string, similar to the waves on a linear array of harmonic oscillators considered in the previous chapter. However, considering the waves along a finite length of string, vibration is made of countless "round trips" between both ends of a string. Therefore, resonance can be found to occur at not only a single frequency but many ones. Finally, this chapter discusses the expression of free vibration as superposition of eigenmodes (or modal functions) with harmonic frequencies as well as graphical representation of wave propagation. The results are physical basis for mathematical representation of a periodic function by Fourier series expansion.

5.1 Wave Propagation of Infinitely Long String

Suppose a long string has density $\rho_0(\text{kg/m})$ for a unit length and its tension is $F_T(\text{N})$. Assign coordinate axis of x along the string, and also assign y for the perpendicular direction to x . A wave along the string $u(x, t)$ is a **transversal wave** in which vibration is perpendicular to the wave propagation. The initial disturbance is propagated as transversal waves on the string with definite speed.

5.1.1 Speed of Transversal Wave

Restoring force due to elasticity of a medium is necessary to excite wave propagation in a medium. The y -component for tension of a string as shown in Fig. 5.1 yields restoring force. Potential energy can be estimated as work done that is necessary to stretch a string against the restoring force. According to the figure, stretch of a string is estimated as [4]

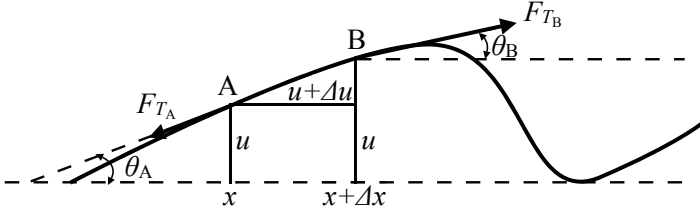


Fig. 5.1 Deformation of string and its tension from [4] Fig. 2.1

$$\sqrt{(\Delta x)^2 + (\Delta u)^2} - \Delta x \cong \frac{1}{2} \left(\frac{\Delta u}{\Delta x} \right)^2 \Delta x \quad (\text{m}) \quad (5.1)$$

for a small interval of the string shown by Δx . Therefore, the work done can be written as

$$E_P = \frac{1}{2} F_T \left(\frac{\Delta u}{\Delta x} \right)^2 \Delta x, \quad (\text{J}) \quad (5.2)$$

where F_T denotes the tension (N). Similarly, kinetic energy can be expressed as

$$E_K = \frac{1}{2} \rho_0 \Delta x v^2, \quad (\text{J}) \quad (5.3)$$

where v denotes vibration velocity (m/s) along the y -axis such that $v = \Delta u / \Delta t$.

If both energies are set to be equal, following the energy preservation law,

$$\frac{(\Delta u / \Delta t)^2}{(\Delta u / \Delta x)^2} = \frac{F_T}{\rho_0} = c^2 \quad (\text{m/s})^2 \quad (5.4)$$

is obtained showing the square of transversal wave speed along a string. Transversal wave speed on a string can be written as the ratio of vibration velocity ($\Delta u / \Delta t$) and slope of displacement with respect to x ($\Delta u / \Delta x$). Here y shows displacement perpendicular to the x -direction along a string of interest. If the slope is the same, the vibration velocity increases with the wave speed, given by the ratio of the tension and density of a string. Wave speed decreases as the density increases under a constant tension, while it increases when the tension is strong under a constant density.

5.1.2 Equation of Wave Propagation on String

A wave traveling on a string is governed by the wave equation in one-dimensional space. Suppose that transverse vibration with small displacement (deflection) travels a long string with the uniform density ρ_0 (kg/m). Recall Fig. 5.1, and assume that the angle θ is small so that $\sin \theta \cong \theta$. The tension that acts on a small portion Δx of the string can then be written as

$$F_{T_B} \sin \theta_B - F_{T_A} \sin \theta_A \cong F_T (\theta_B - \theta_A) \quad (5.5)$$

$$\theta_B - \theta_A \cong \frac{\partial \theta(x)}{\partial x} \Delta x \quad (5.6)$$

$$\tan \theta(x) \cong \theta(x) \cong \frac{\partial u(x,t)}{\partial x} \quad (5.7)$$

where $u(x,t)$ denotes the deflection of the string at the small portion. Then the equation of motion for the small portion of the string becomes

$$\rho_0 \Delta x \frac{\partial^2 u(x,t)}{\partial t^2} = F_T \Delta x \frac{\partial^2 u(x,t)}{\partial x^2}; \quad (5.8)$$

namely,

$$\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2} \quad (5.9)$$

is obtained as the **wave equation for transversal waves on a string** where

$$c^2 = \frac{F_T}{\rho_0} \quad (5.10)$$

and c (m/s) is the speed of the waves on a string. Vibration of a string is an important example of wave propagation along a one-dimensional axis.

5.1.3 Initial Displacement and Its Propagation

A progressive wave can be interpreted as free vibration of an oscillator, but its frequency is arbitrary. External force to create initial disturbance, from which progressive waves are generated, can be characterized in terms of initial conditions. Suppose that a portion of a string is picked up and released quietly as shown in Fig. 5.2. This is a simplified example for an initial condition of string vibration such as in a harp. The displacement is equally separated into two parts, both of which are in the shape of the initial disturbance. As the peak of the central part of the initial displacement decreases, two equally divided progressive waves are propagated to the right and left, respectively.

Given the initial displacement and velocity (initial condition) such that

$$u(x,0) = a(x) \quad (\text{m}) \quad (5.11)$$

$$v(x,0) = 0, \quad (\text{m/s}) \quad (5.12)$$

the initial displacement is propagated to the left and right separately such that

$$u(x,t) = \frac{1}{2} (a(x+ct) + a(x-ct)) \quad (\text{m}) \quad (5.13)$$

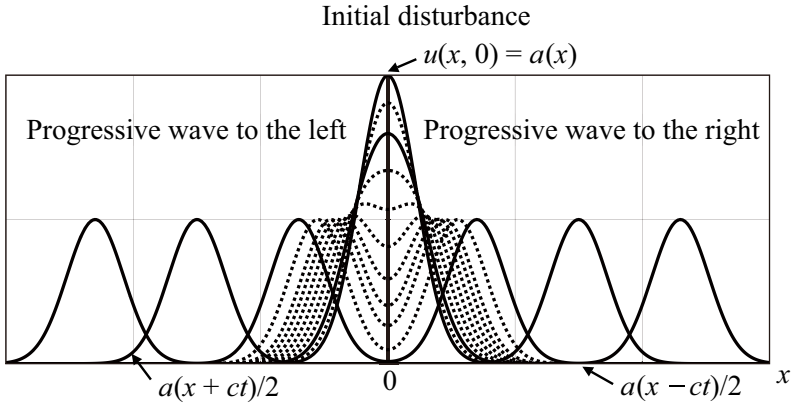


Fig. 5.2 Initial displacement of string and propagation waves from [9] (Fig.5.1)

after the string is quietly released. Here the first term represents the progressive wave to the left (negative side of x -direction), and the second one represents a wave in the opposite direction. The waves are propagated without change in their forms as time passes.

5.1.4 Propagation of Initial Velocity

Suppose that initial velocity is given to a portion of a string as shown in Fig. 5.3 instead of the example above. It represents a simplified initial condition for string vibration such as in a piano. The condition is expressed as

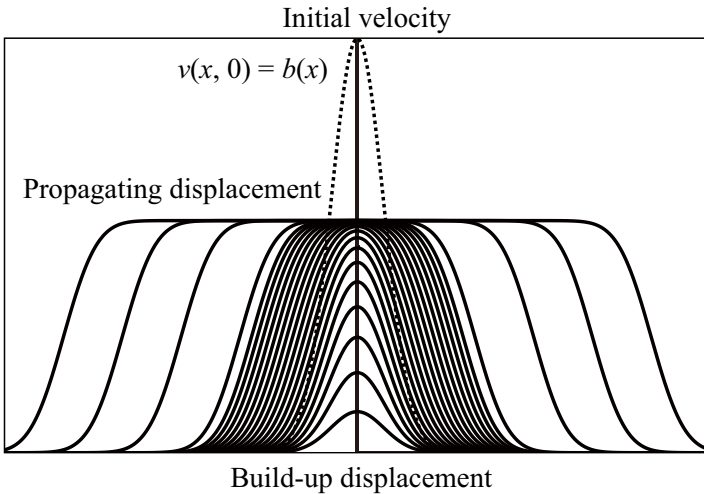


Fig. 5.3 Initial velocity of string and propagation waves from [9] (Fig.5.2)

$$u(x, 0) = 0 \quad (\text{m}) \quad (5.14)$$

$$v(x, 0) = b(x). \quad (\text{m/s}) \quad (5.15)$$

The initial velocity is also equally separated into two parts; however, displacement excited by the initial velocity is different. It is expressed as

$$u(x, t) = \left(\frac{1}{2ct} \int_{x-ct}^{x+ct} b(z) dz \right) t, \quad (\text{m}) \quad (5.16)$$

which shows that displacement is propagated according to the average of the initial velocity between $x - ct$ and $x + ct$.

Displacement at the position of $x = 0$, the center to which initial velocity is given as shown by the broken line in the figure, increases up as time passes. Similarly, the build-up process at position x is expressed as a product of the time and average of the initial velocity between $x - ct$ and $x + ct$. The interval over which the average is taken increases as time passes; this indicates that the area in which a progressive wave travels spreads as time passes. The spreading speed of the area indicates the wave speed. The displacement converted from the initial velocity does not disappear as pointed out in the previous chapter, even after the progressive wave passes.

5.1.5 Generalized Initial Conditions and Propagation of Waves

Initial conditions can be expressed by both displacement and velocity. This means propagating waves on a string can be predicted by the initial displacement and velocity. If the initial conditions are given such that

$$u(x, 0) = a(x) \quad \text{for displacement} \quad (\text{m}) \quad (5.17)$$

$$v(x, 0) = b(x) \quad \text{for velocity,} \quad (\text{m/s}) \quad (5.18)$$

then formal expressions of progressive waves of displacement and velocity

$$u(x, t) = \frac{1}{2} (a(x + ct) + a(x - ct)) + \left(\frac{1}{2ct} \int_{x-ct}^{x+ct} b(z) dz \right) t \quad (\text{m}) \quad (5.19)$$

$$v(x, t) = \frac{c}{2} (a'(x + ct) - a'(x - ct)) + \frac{1}{2} (b(x + ct) + b(x - ct)), \quad (\text{m/s}) \quad (5.20)$$

can be obtained, respectively. Displacement (or velocity) is a superposition of the propagating initial displacement (velocity) and converted from the initial velocity (displacement). Therefore, it is possible that superpositions of waves might cancel each other out, and consequently a one-way traveling wave is possible. Figure 5.4 is an example of such a one-way wave propagation, where the initial conditions are given as [18]

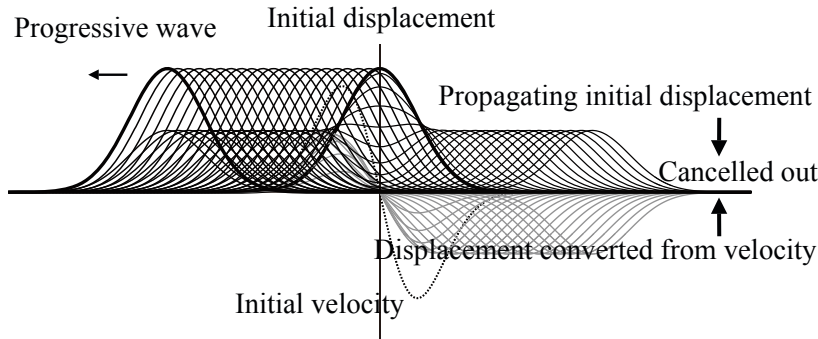


Fig. 5.4 Example of one-way propagating wave from [9] (Fig.5.3)

$$u(x,t) = a(x) \quad \text{for displacement} \quad (\text{m}) \quad (5.21)$$

$$v(x,t) = c \cdot a'(x) \quad \text{for velocity.} \quad (\text{m/s}) \quad (5.22)$$

A one-way progressive wave such as

$$u(x,t) = \frac{1}{2} (a(x+ct) + a(x-ct)) + \left(\frac{1}{2ct} \int_{x-ct}^{x+ct} b(z) dz \right) t = a(x+ct) \quad (5.23)$$

has been confirmed possible due to cancellation between the initial and converted displacements.

5.2 Boundary Conditions and Harmonic Vibration

The previous section described progressive waves along an infinitely long string caused by initial disturbance to the string. However, in a practical situation, a string normally has a finite length. Therefore, a progressive wave will eventually come to the ends of a string of interest. At such a boundary, a progressive (incident) wave returns as a reflection wave. Consequently, propagation of the initial disturbance without change of the waveform is no longer expected. Thus, this section describes the normal modes of vibration that are composed of a fundamental and its harmonics.

5.2.1 Wave Reflection at Boundary

Consider a string with one end clamped as shown in Fig. 5.5, where the endpoint (boundary) is set at $x = 0$. A clamped boundary at $x = 0$ specifies a condition at a boundary (or a boundary condition) such that the string does not move at the boundary. That is, displacement of a string can be set at the boundary such that

$$u(0,t) = 0. \quad (\text{m}) \quad (5.24)$$

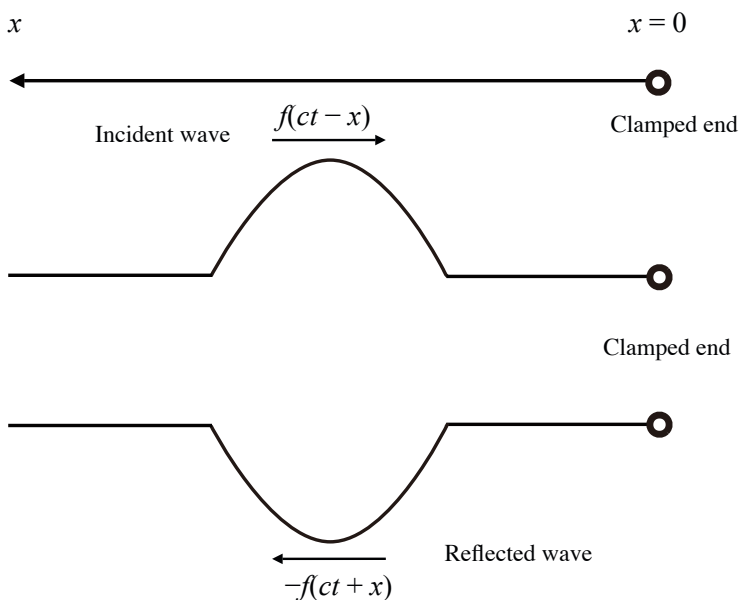


Fig. 5.5 Progressive wave reflected at clamped end ($x = 0$) of string

The boundary condition above regulates a progressive wave that approaches boundary from the negative side of the string as shown in Fig. 5.5, so the displacement might be cancelled at the boundary. The boundary condition or the regulation needs another wave that cancels the coming wave at the boundary. The other wave is called a **reflection wave**, which returns to the left in the figure.

Take an approaching progressive wave as

$$u_1(x, t) = g(ct - x), \quad (5.25)$$

and set an opposite (returning) progressive wave to be

$$u_2(x, t) = f(ct + x). \quad (5.26)$$

By superposition of these two waves such that

$$u(x, t) = u_1(x, t) + u_2(x, t) = g(ct - x) + f(ct + x), \quad (5.27)$$

the reflective wave is obtained such that

$$f(ct + x) = -g(ct - x) \quad (5.28)$$

following the boundary condition.

Note here that there are two ways of representing a progressive wave: $g(ct - x)$ or $g(x - ct)$. There is no significant difference between these two. The difference

exists in taking a "reference" for representation of time or spatial distance. When a progressive wave is expressed as $g(ct - x)$, the wave is represented based on a "sense of time", such as how long the wave will take to arrive at a position x . That is, the origin $x = 0$ is always taken as a reference. In contrast, a progressive wave in the form $g(x - ct)$ follows the sense of how long ago the wave left the origin $x = 0$. An observation point at x is taken as a reference in this case. Both ways are used to represent progressive waves in this book.

5.2.2 Vibration Patterns of Finite Length of String

Suppose that a finite length of string is clamped at both ends, denoted by $x = 0$ or $x = L$. Here $L(\text{m})$ is the length of the string. It seems natural to suppose a pattern of vibration as shown in Fig. 5.6($n = 1$). This pattern meets the boundary conditions at both ends and is called a **fundamental mode** of free vibration of a finite length of string. With the assumption that progressive waves simultaneously leave the center of the string at $t = 0$ and respectively head to the right and left, both waves meet again at the center after traveling back and forth between both ends of the string.

The travel of waves on a string is **periodic**. A **period** for the back and forth travel of a wave is given by $2L/c$ (s), where c denotes the wave speed (m/s). Therefore, a period $T_1 = 2L/c$ (s) is called the **fundamental period**, and its inverse, i.e., $\nu_1 = c/2L$ (Hz), is called the **fundamental**. The fundamental is interpreted as the

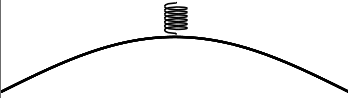
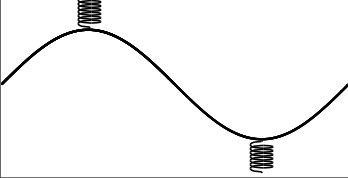
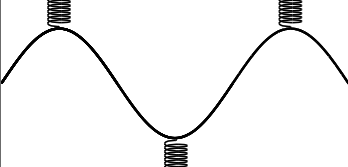
Order	Vibration pattern	Wavelength: λ_n	Resonant freq: ν_n
$n = 1$		$\lambda_1 = 2L$	$\nu_1 = c/2L$
$n = 2$		$\lambda_2 = L$	$\nu_2 = 2\nu_1$
$n = 3$		$\lambda_3 = 2L/3$	$\nu_3 = 3\nu_1$

Fig. 5.6 Vibration patterns for finite length of string with clamped ends from [19] (Fig.2.34)

lowest eigenfrequency of a finite length of string, and thus there are countless eigenfrequency for vibration of a finite length of string as described below.

The pattern of vibration shown in Fig. 5.6 ($n = 1$) is periodic with a period of $2L/c$. A vibration pattern does not necessarily have to be the pattern stated above, however. Figure 5.6 ($n = 2$) is another pattern of vibration. By equal division of a string into two parts, the "fundamental for each part" is given by c/L . Thus if the divided segments are smoothly reconnected, a vibration pattern could be obtained as shown in Fig. 5.6 ($n = 2$). Here the fundamental for each of the divided parts, i.e., $v_2 = 2v_1 = c/L$ (Hz), is called the **2nd harmonics**, i.e., the second eigenfrequency. There is a dividing point, which is called a **node**, where the displacement is null as if that point were a virtual boundary (end). On the other hand, the position on a string where displacement takes its maximum is called a **loop**.

Note however that another condition is needed for connecting the divided parts; namely, vibration patterns are in anti-phase with respect to the connecting point (virtual boundary) as implied by smooth connection. By repetition of this dividing and connecting process, the n -th harmonics (eigenfrequency) whose frequency is given by $v_n = nv_1$ can be obtained. Figure 5.6 ($n = 3$) shows the 3rd harmonics. Vibration patterns like n -th harmonics are called **normal modes** of vibration.

There are $n - 1$ nodes in the n -th mode in general. Any pair of adjacent parts with respect to the node makes vibrations that are in anti-phase to each other. In other words, vibration reverses its phase at every node. As a result, the vibration patterns suggest that free vibration of a finite length of string might be represented by odd and periodic functions. Here the term "periodic" has a double meaning; namely, the functions are periodic with respect to time with a period of $2L/c$ (s), and in addition, that the functions are also spatially periodic with an interval of $2L$.

Now consider free vibration of a finite string whose length is L (m) such that

$$u(x, t) = f(x + ct) + g(x - ct) \quad (\text{m}) \quad (5.29)$$

under the initial conditions for the displacement and velocity such as

$$u(x, 0) = u_0(x) = a(x) \quad 0 \leq x \leq L \quad (\text{m}) \quad (5.30)$$

$$v(x, 0) = 0. \quad (\text{m/s}) \quad (5.31)$$

By following the general solution for an infinitely long string derived in the previous section, the displacement can be written as

$$u(x, t) = \frac{1}{2}(u_0(x + ct) + u_0(x - ct)). \quad (\text{m}) \quad (5.32)$$

Here, recalling the boundary conditions for a clamped string at $x = 0$ and $x = L$ such that

$$u(0, t) = \frac{1}{2}(u_0(ct) + u_0(-ct)) = 0 \quad (5.33)$$

$$u(L, t) = \frac{1}{2}(u_0(L + ct) + u_0(L - ct)) = 0, \quad (\text{m}) \quad (5.34)$$

the relation

$$u_0(-ct) = -u_0(ct) \quad (m) \quad (5.35)$$

$$u_0(L - ct) = -u_0(L + ct) \quad (m) \quad (5.36)$$

can be obtained, and thus

$$u_0(L + ct) = -u_0(L - ct) = u_0(ct - L) = u_0(L + ct - 2L) \quad (m) \quad (5.37)$$

holds well. Consequently, it can be confirmed that functions which represent the propagating waves are odd and are spatially periodic with an interval of $2L$.

5.2.3 Generalized Boundary Conditions: End Correction

Boundary conditions for vibration of a string can be generalized to represent a more realistic situation. This type of modification of the boundary conditions is well known as a historical issue about end correction for an acoustic tube[7]. If the end of a tube is "open" where there is no acoustic pressure, then no sound is radiated from the tube. In addition, the fundamental of an open tube is different from that estimated by $c/2L$. Namely, the "effective length" of the tube must be modified from the geometrical length of the tube, in general, to appropriately estimate the fundamental. This is due to the fact that the end of a tube seems to be open but that from an acoustical point of view it is not. This is a very important issue in designing musical instruments like organs.

It is also an important issue with regard to the waves travelling on a string in a musical instrument such as a piano[20]. If the string of a piano is perfectly fixed at the end points, then no vibration is transmitted to the sound board of the piano, and consequently no sound is radiated. The boundary conditions can be formulated in terms of the **impedance** of vibration[21].

Suppose that a string is terminated at $x = 0$ by **mechanical impedance** Z_M as shown in Fig. 5.7. Here the mechanical impedance Z_M denotes a ratio between the

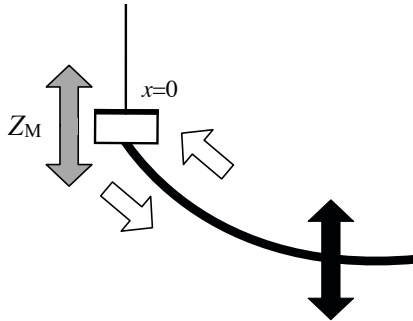


Fig. 5.7 End of string at $x = 0$ that is terminated by mechanical impedance Z_M

force and velocity at the point of interest. Recall that deflection travels on a string as sinusoidal waves by discarding the time components such that

$$U(x) = Ae^{-ikx} + Be^{ikx}, \quad (5.38)$$

where k denotes the **wavenumber** (1/m) defined by $k = \omega/c$. Here the first term represents the **incident wave** and the second one shows the **reflection wave** from the boundary. Thus, the ratio B/A expresses the **reflection coefficient from the boundary**, which depends on the boundary condition.

The deflection waves described above must meet the boundary condition at $x = 0$. From Fig. 5.1, the force that acts on the boundary can be expressed as

$$F(x)|_{x=0} = -F_T \sin \theta \cong -F_T \theta \cong -F_T \left. \frac{\partial U(x)}{\partial x} \right|_{x=0}, \quad (5.39)$$

where force in the upper direction is indicated with a positive sign. According to the boundary condition, namely the boundary impedance, the velocity $V(x)$ can be expressed as

$$F(x)|_{x=0} = -F_T \left. \frac{\partial U(x)}{\partial x} \right|_{x=0} = Z_M V(x) \Big|_{x=0}. \quad (5.40)$$

Therefore, by introducing the velocity and force at the boundary such that

$$F(0) = ikF_T(A - B) \quad (\text{N}) \quad (5.41)$$

$$V(0) = i\omega(A + B) \quad (\text{m/s}), \quad (5.42)$$

the relationship

$$F_T(A - B) = Z_M c(A + B) \quad (5.43)$$

holds well between the force and velocity at the boundary, where c (m/s) denotes the speed of waves on the string. Consequently, the reflection coefficient B/A is obtained as

$$\frac{B}{A} = -\frac{\hat{Z}_{Mc} - 1}{\hat{Z}_{Mc} + 1} \quad (5.44)$$

in terms of the boundary impedance, where $\hat{Z}_M = Z_M/F_T$.

The boundary impedance can be tentatively represented in terms of the mass, spring constant, or friction coefficient based on vibration of a single degree of freedom of the system. In other words, it can be interpreted as the vibrating string being connected to the rigid boundary via a mass-spring system. Suppose that the boundary impedance can be represented by the mass so that the force is in proportion to the acceleration, or the impedance can be expressed as a pure imaginary number. Then assuming the impedance can be written as

$$\hat{Z}_{Mc} = i \cot k y_{Ec} \quad (y_{Ec} > 0), \quad (5.45)$$

the reflection coefficient can be rewritten as

$$\frac{B}{A} = -\frac{i \cot k y_{Ec} - 1}{i \cot k y_{Ec} + 1} = -e^{2ik y_{Ec}}. \quad (5.46)$$

After substituting this reflection coefficient into Eq. 5.38,

$$U(x) = A e^{-ikx} + B e^{ikx} = A e^{iky_{Ec}} \left(e^{-ik(x+y_{Ec})} - e^{ik(x+y_{Ec})} \right) \quad (5.47)$$

is derived. This outcome is as if the length of string were shortened by the amount of y_{Ec} .

In contrast, if the boundary impedance is a value such that the force is in proportion to the deflection of the string, i.e.,

$$\hat{Z}_{Mc} = -i \cot k y_{Ec} \quad (y > 0), \quad (5.48)$$

then the reflection coefficient becomes

$$\frac{B}{A} = -e^{-2iky_{Ec}}. \quad (5.49)$$

Therefore, the deflection wave can be rewritten as

$$U(x) = e^{-iky_{Ec}} \left(e^{-ik(x-y_{Ec})} - e^{ik(x-y_{Ec})} \right). \quad (5.50)$$

This expression can be visualized as if the length of string were extended by y_{Ec} .

If the impedance is regarded as resistive, namely the force is in proportion to the vibration velocity, then the impedance is a real function. Therefore, the reflection coefficient is also real with a negative sign whose magnitude is smaller than unity. The results can be intuitively interpreted[22]. For the case of massive impedance, the phase of the reflection wave seems to "lead" the incident wave by $2ky_{Ec}$ at the boundary. This can be interpreted as if the reflection wave is in anti-phase to the incident wave at $x = -2y_{Ec}$, and thus the two waves meet at $x = -y_{Ec}$ so that the "node" might be created by cancellation. This node makes the length of the string effectively shorter by y_{Ec} .

In contrast, for the stiffness boundary, the phase of the reflection wave is delayed by $2ky_{Ec}$ from the incident wave at $x = 0$. This can be understood as the reflection from the virtual end at $x = y_{Ec}$. Namely, the string becomes virtually longer than its geometrical length.

Only for the case of resistive impedance is the energy of the incidental wave consumed, i.e., the vibration of a string is transmitted across the boundary. Therefore, the magnitude of the reflective wave is smaller than that of the incidental wave. In a real situation, the boundary impedance can be interpreted as a mixture of the three components.

5.2.4 Effects of Boundary Conditions on Eigenfrequencies

The end correction for a one-dimensional vibrating system such as a vibrating string was described according to the wave theory in the previous section. However, such a boundary effect on the eigenfrequencies can be intuitively understood in terms of vibration for two-degree-of-freedom systems. Namely, it can be described from the view point of transmission or reduction of vibration between vibrating systems.

Consider a coupled vibrating system as shown in Fig. 5.8. Set the exciting force $F_X e^{i\omega t}$ for the mass 2. Then, two equations of motion are written as

$$M_1 \frac{d^2 x_1(t)}{dt^2} = -K_1 x_1(t) - K_2 (x_1(t) - x_2(t)) \quad (5.51)$$

$$M_2 \frac{d^2 x_2(t)}{dt^2} = -K_2 (x_2(t) - x_1(t)) + F_X e^{i\omega t}. \quad (5.52)$$

By substituting $x_1(t) = U_1(\omega) e^{i\omega t}$ and $x_2(t) = U_2(\omega) e^{i\omega t}$ for the equation above,

$$U_1(\omega) = \frac{K_2 F_X}{(K_1 + K_2 - \omega^2 M_1)(K_2 - \omega^2 M_2) - K_2^2} = \frac{K_2}{D(\omega)} F_X \quad (5.53)$$

$$U_2(\omega) = \frac{K_1 + K_2 - \omega^2 M_1}{D(\omega)} F_X \quad (5.54)$$

are derived. Here system 1 in the figure can be regarded as the system that represents the boundary condition. In other words, if the boundary can be regarded as a completely fixed point, then the vibrating system can be simplified to the single-degree-of-freedom system only.

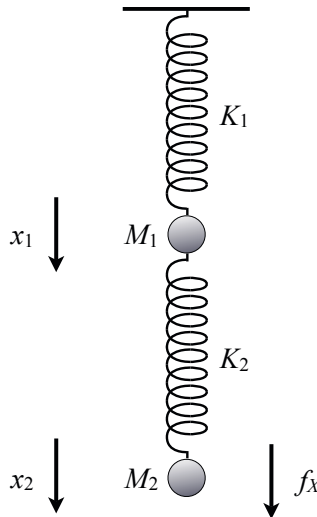


Fig. 5.8 Example of coupled vibrating system

It might be interesting to see the eigenfrequencies of the system including the effect of the "boundary condition". For that purpose, first suppose that the boundary can be assumed to be purely elastic without the mass effect. By assuming such a boundary condition, the denominator of Eq. 5.54 can be written as

$$D(\omega) \cong K_1 K_2 - (K_1 + K_2) \omega^2 M_2 \quad (5.55)$$

according to the assumption $M_1 \rightarrow 0$. Thus the eigenfrequency can be obtained as

$$\omega^2 = \frac{K_2}{M_2} \frac{K_1}{K_1 + K_2} < \frac{K_2}{M_2}. \quad (5.56)$$

Consequently, it can be seen that the eigenfrequency becomes lower than the original one due to the elastic boundary. This outcome corresponds to the end correction of the elastic boundary for a finite length of a string.

On the other hand, assuming a massive boundary, namely $K_1 \rightarrow 0$, then

$$D(\omega) \cong \omega^2 (M_1 M_2 \omega^2 - K_2 (M_1 + M_2)). \quad (5.57)$$

Thus the eigenfrequency becomes

$$\omega^2 = \frac{K_2}{M_2} \frac{M_1 + M_2}{M_1} > \frac{K_2}{M_2}. \quad (5.58)$$

The outcome above represents the eigenfrequency becoming higher than the original because of the massive boundary effect. This also corresponds to end correction for the vibration of a finite length string.

5.3 Driving Point Impedance of Vibrating String

Excitation of vibration motion and its travel in a medium can be represented in terms of impedance. The **driving point impedance** is defined by the ratio of the force to vibration velocity at the exciting position.

5.3.1 Driving Point Impedance of Travelling Wave on String

Assume a sinusoidal wave, excited at one end $x = 0$ and travelling on a string, whose displacement $u(x, t)$ is written as

$$u(x, t) = U(x) e^{i\omega t} = A e^{i(\omega t - kx)}. \quad (m) \quad (5.59)$$

Recall that the external force applied at $x = 0$ can be expressed by Eq. 5.39[21]. Thus the displacement and its velocity can be rewritten as

$$u(x, t) = \frac{F_X}{ikF_T} e^{i(\omega t - kx)} = \frac{F_X}{i\omega \rho_0 c} e^{i(\omega t - kx)} \quad (m) \quad (5.60)$$

$$v(x, t) = \frac{F_X}{\rho_0 c} e^{i(\omega t - kx)}, \quad (\text{m/s}) \quad (5.61)$$

where $F_T = \rho_0 c^2$ (N) is the tension, ρ_0 (kg/m) denotes the mass of the string for a unit length, and c (m/s) is the wave speed. Consequently, the **driving point mechanical impedance** of the string becomes

$$Z_{M_{\text{in}}} = \left. \frac{f_X}{v} \right|_{x=0} = \rho_0 c. \quad (\text{N} \cdot \text{s/m}) \quad (5.62)$$

It can be seen that the driving point impedance is real, so a product of the force and velocity is also real such that

$$W_0 = \frac{1}{2} \text{Re}(f_X^* v) \Big|_{x=0} = \frac{1}{2} \text{Re} \frac{|F_X|^2}{Z_{M_{\text{in}}}} = \frac{1}{2} \frac{|F_X|^2}{\rho_0 c} \quad (\text{W}) \quad (5.63)$$

on average over a unit time interval, where $\text{Re}(\cdot)$ denotes taking the real part of the corresponding complex quantity, f_X (N) denotes the sinusoidal external source, f_X^* means the complex conjugate of f_X , and $1/2$ corresponds to the time-average for a squared sinusoidal function. This means the energy continuously travels from the source into the string.

5.3.2 Driving Point Impedance of Standing Wave on Finite Length of String

Instead of the example above, suppose that there is a finite length of string whose ends are fixed at $x = 0$ and $x = L$. Now set the external sinusoidal source at $x = 0$; the waves travelling on the string must meet the boundary and source conditions. Take the two travelling waves that cause the displacement of vibration such that

$$u(x, t) = A e^{i(\omega t - kx)} + B e^{i(\omega t + kx)}. \quad (\text{m}) \quad (5.64)$$

For the boundary and source conditions given by

$$u(L, t) = 0 \quad (5.65)$$

$$-F_T \left. \frac{\partial u(x, t)}{\partial x} \right|_{x=0} = F_X e^{i\omega t} = f_X(0, t), \quad (5.66)$$

$$A e^{-ikL} + B e^{ikL} = 0 \quad (5.67)$$

$$ikF_T A - ikF_T B = F_X \quad (5.68)$$

hold well.

By solving the simultaneous equations above,

$$A = \frac{F_X e^{ikL}}{i2\rho_0 c \omega \cos kL} \quad (5.69)$$

$$B = \frac{-F_X e^{-ikL}}{i2\rho_0 c \omega \cos kL} \quad (5.70)$$

are obtained as the solutions. After substituting A and B above into Eq. 5.64, the displacement of vibration can be written as

$$u(x,t) = \frac{1}{\rho_0 c \omega} \frac{\sin k(L-x)}{\cos kL} f_X(0,t). \quad (\text{m}) \quad (5.71)$$

and its velocity is

$$v(x,t) = \frac{\partial u(x,t)}{\partial t} = \frac{i}{\rho_0 c} \frac{\sin k(L-x)}{\cos kL} f_X(0,t). \quad (\text{m/s}) \quad (5.72)$$

Therefore, the driving point mechanical impedance becomes

$$Z_{M_{in}} = \frac{f_X(0,t)}{v(0,t)} = -i\rho_0 c \cot kL. \quad (\text{N} \cdot \text{s/m}) \quad (5.73)$$

Note here that driving point mechanical impedance is purely imaginary. This fact indicates that the power might not be absorbed by the travelling waves on the string. This is an important difference between the **travelling** and **standing** waves. It is the outcome due to the "un-realistic" assumption that there is no damping in the vibration of a string. Namely, the boundary at $x = L$ is ideally fixed without any motion.

The results showing no energy transfer imply that it is impossible to define which is the external source of force for the motion of vibration. Namely, if the system were at the steady state and external source stopped, the motion of the system and its external piston-like source would continue forever. The energy goes back and forth between the piston and the string.

The situation above is not possible in a practical case, and it is only possible from an ideal point of view assuming no energy consumption. Recall that strings are pinned at the ends on the sound board in a piano. If no energy is transferred from the vibrating strings into the sound board, no sound is available from the piano. Such energy transfer through the boundaries is described in terms of the **end correction** for the boundaries as described in subsection 5.2.3.

5.3.3 Driving Point Impedance and Power Injection from External Source

Now reconsider power injection from the external source into a system of interest subject to a small amount of energy loss being assumed in the system. Recall the driving point mechanical impedance $Z_{M_{in}}$ at $x = 0$ of a finite length of string

described in the previous subsection. The power transmitted from the source into the string was written by Eq. 5.73. Here, by introducing the complex variable

$$k = k_r - i\beta \quad (k_r \gg \beta), \quad (5.74)$$

the driving point impedance can be written as

$$Z_{M_{in}}(0) = -i\rho_0 c \frac{\cos(k_r - i\beta)L}{\sin(k_r - i\beta)L} \cong -i\rho_0 c \frac{\cos k_r L + i\beta L \sin k_r L}{\sin k_r L - i\beta L \cos k_r L}. \quad (\text{N} \cdot \text{s/m}) \quad (5.75)$$

Therefore, the power injected from the source becomes

$$W_X \cong \frac{1}{2} \frac{|F_X|^2}{\rho_0 c} \frac{\beta L}{\cos^2 k_r L} \cong \frac{1}{2} \frac{|F_X|^2}{\rho_0 c} \frac{\beta L}{\cos^2 k L} \quad (\text{W}) \quad (5.76)$$

where $k \cong k_r$ is assumed.

Now recall the vibration displacement $u(x, t)$ given by Eq. 5.71 depends on the observation point of x . By taking the spatial average of the squared magnitude response over all the observation points on the string,

$$\langle u^2 \rangle = \frac{1}{2} \overline{u(x, t) \cdot u^*(x, t)} = \frac{1}{2} \frac{1}{L} \int_0^L |u(x, t)|^2 dx = \frac{1}{4} \frac{|F_X|^2}{\rho_0 c} \frac{1}{\cos^2 k L} \frac{1}{\rho_0 c \omega^2}, \quad (\text{m}^2) \quad (5.77)$$

is obtained. The following relationship

$$W_X = \langle u^2 \rangle \cdot \rho_0 c \omega^2 \cdot 2\beta L \quad (\text{W}) \quad (5.78)$$

is found to hold well between the injected power and squared magnitude response averaged over all the space. It can be interpreted as the **energy balance equation** between the injected and consumed power in the system. In other words, the power response from the source located in the system can be estimated by the spatial average of the squared quantity of the vibration magnitude.

5.3.4 Driving Point Impedance and Source Position

The driving point mechanical impedance depends on the source position for standing waves on a finite length of string. Suppose that the external source is located at $x = x' \neq 0$. Set the mechanical impedance for the right hand side of the string to be $Z_{MR_{in}}$. By introducing the velocity at the source position $v(x', t)$, the force for the right hand side string can be written as

$$f_{X_R}(x', t) = Z_{MR_{in}} v(x', t) = -i\rho_0 c \cot k(L - x') \cdot v(x', t). \quad (\text{N}) \quad (5.79)$$

where $k \cong k_r$ is assumed and L denotes the length of the string. By similarity taking the left hand side of the string,

$$f_{X_L}(x', t) = Z_{ML_{in}} v(x', t) = -i\rho_0 c \cot kx' \cdot v(x', t). \quad (\text{N}) \quad (5.80)$$

is obtained. Consequently, the driven point mechanical impedance is given by

$$Z_{M_{in}}(x') = \frac{f_{X_R}(x', t) + f_{X_L}(x', t)}{v(x', t)} = -i\rho_0 c \frac{\sin kL}{\sin kx' \sin k(L-x')} \quad (\text{N} \cdot \text{s/m}) \quad (5.81)$$

where $c^2 = F_T / \rho_0$. As shown here, the driving point impedance actually depends on the location of the source. Note that the driving point impedance does not depend on the type of external source. If a velocity type of external source is applied at $x = x'$ on the string instead, then the impedance is the same as that by the equation above.

5.3.5 Poles and Zeros of Driving Point Impedance

The impedance function can be formulated by **poles** and **zeros**[23][24]. The poles correspond with the singularities such as the zeros of the denominator, while the zeros indicate zeros of the numerator. Recall the driving point impedance

$$Z_{M_{in}}(0) = -i\rho_0 c \frac{\cos kL}{\sin kL} \quad (5.82)$$

$$Z_{M_{in}}(x') = -i\rho_0 c \frac{\sin kL}{\sin kx' \sin k(L-x')} \quad (\text{N} \cdot \text{s/m}) \quad (5.83)$$

It can be confirmed that the poles and zeros are interlaced on the frequency axis[23][24]. Suppose that k is a wavenumber of interest. The number of zeros distributed under the wavenumber can be estimated as

$$N_z = \frac{k}{\pi/L}. \quad (5.84)$$

By similarity considering the poles, those numbers can be also estimated as

$$N_p = N_{p_1} + N_{p_2} \quad (5.85)$$

$$N_{p_1} = \frac{k}{\pi/x'} \quad (5.86)$$

$$N_{p_2} = \frac{k}{\pi/(L-x')} \quad (5.87)$$

Therefore, the difference in the numbers of the poles and zeros is

$$N_z - N_{p_1} - N_{p_2} = \frac{k}{\pi} ((L-x') - (L-x')) = 0. \quad (5.88)$$

Namely, it can be understood that the poles and zeros must be interlaced.

The poles and zeros of driving point mechanical impedance are closely related to the eigenfrequencies of the system. Suppose that the external source can be assumed as **constant force**. The excited velocity on the string by the force can be written as

$$v(x', t) = \frac{f_X(x', t)}{Z_{M_{in}}(x')} \quad (\text{m/s}) \quad (5.89)$$

in terms of the driving point mechanical impedance. Then the **zeros of the impedance** create the **resonance for the standing waves** of corresponding frequencies with the zeros, while the **poles** create the **anti-resonance** of the waves. Namely, standing waves of the frequencies corresponding to the poles cannot be excited by the force at the source position. On the other hand, the poles create the resonance when the external source is assumed to be **constant velocity**. The external force can be rewritten as

$$f_X(x = x', t) = v(x', t)Z_{M_{in}}(x') \quad (\text{N}) \quad (5.90)$$

according to the driving point mechanical impedance.

In the case of the piano, the external source position (hammer position) is set at around $L/8$ where L is the length of a string in the piano[25]. This is because around 7-9 harmonics might be removed from the higher harmonics. This can be interpreted in terms of driving point impedance. The driving point impedance becomes

$$Z_{M_{in}}(x')|_{x'=L/8} = -i\rho_0 c \frac{\sin kL}{\sin(kL/8)\sin(7kL/8)}. \quad (\text{N} \cdot \text{s/m}) \quad (5.91)$$

Here the first pole is given by

$$\frac{k_{p1}L}{8} = \frac{2\pi v_{p1}L}{8c} = \pi. \quad (5.92)$$

Namely,

$$v_{p1} = 8 \frac{c}{2L} = 8v_1 \quad (\text{Hz}) \quad (5.93)$$

is obtained. The outcome indicates that the standing wave of the frequency given by v_{p1} cannot be excited by the external force, subject to the source being assumed as constant.

5.4 Propagation of Initial Disturbance along Finite Length of String

Propagation of initial disturbance on a finite length of string can also be graphically illustrated according to the boundary conditions. The notion of normal modes or harmonics cannot be explicitly seen in the graphical method for representing wave propagation. However, it can be found that such a wave theoretic concept as normal modes must be hidden in the graphical method by recalling that periodically propagating waves are expressed as odd and spatially periodic functions following the boundary conditions.

5.4.1 Propagation of Initial Displacement

Consider a finite length of string with initial displacement as shown in Fig. 5.9. The initial displacement in the interval between $x = 0$ and $x = L$ can be extended as an odd and periodic function.

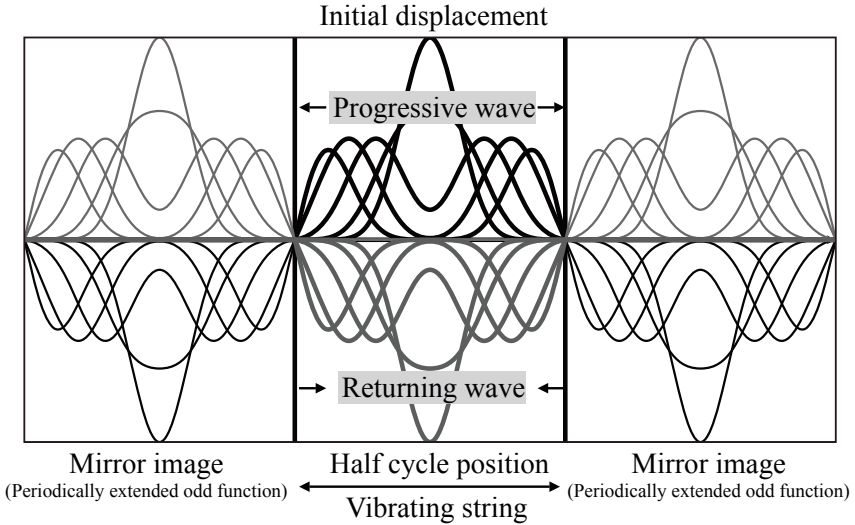


Fig. 5.9 Propagation of initial displacement for finite length of string with clamped ends from [9] (Fig.5.6)

Similar to wave propagation along a long string as described in section 5.1, the initial disturbance is equally separated and propagated to the left and right even on a finite length of string. However, for a finite length of string, such separation and propagation simultaneously occur in every interval between $x = lL$ and $x = (l+1)L$ where l is an integer, as if the disturbance occurred in every interval on an infinitely extending virtual string. Here, every wave entering the interval ($x = 0$ to $x = L$) from outside is interpreted as a reflection wave. It takes a long time for a reflection wave to return to the original interval, as a virtually extended interval from which the reflection comes is far from the original. In other words, a reflection wave coming from the interval between $x = lL$ and $x = (l+1)L$ represents an l -th reflection wave after l collisions into the boundary.

Consequently, waves that are propagated on the string between $x = 0$ and $x = L$ are obtained by periodic superposition of all the reflection waves from outside the interval. The period of vibration is $T_1 = 2L/c = 1/v_1$ (s), which actually corresponds to the time interval of successive reflections; namely, reflection waves enter every interval of T_1 .

5.4.2 Propagation of Initial Velocity

Figure 5.10 illustrates propagation of displacement due to the initial disturbance of velocity instead of displacement. Vibration including build-up and decaying processes is cyclic with a period of $2L/c$ (s), and the vibration pattern itself is extended as an odd and spatially periodic function with an interval of $2L$ (m). Displacement shown in the figure was due to conversion from the initial velocity because of no initial displacement. The displacement might be interpreted as left over after a wave passed in every cycle.

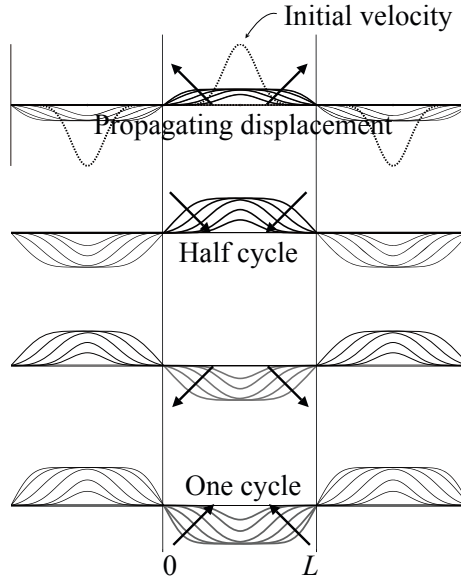


Fig. 5.10 Propagation of displacement due to initial velocity for finite length of string with clamped ends from [9] (Fig.5.7)

5.5 Impulse Response and Its Transfer Function for Vibrating String

A vibrating string is a good example that shows the relationship between an impulse response and its transfer function for a linear system.

5.5.1 Impulse Response of Finite Length of String

Suppose that a finite length of string is excited by an impulse-like initial excitation as position S as shown by Fig. 5.11. The response observed at position O is composed of four types of pulse series that start at different initial positions on the time axis, but with the same time intervals of successive pulses as shown in Fig. 5.12.

Here set the time intervals as follows:

$$\tau_{SR} = (x - x')/c = N_d T_s \quad (5.94)$$

$$\tau_{SL} = (2x' + (x - x'))/c = N_b T_s \quad (5.95)$$

$$\tau_{OR} = 2x_o/c = N_\eta T_s \quad (5.96)$$

$$\tau_{OL} = 2(x)/c = N_\gamma T_s \quad (5.97)$$

and

$$\tau_{SR} + \tau_{OR} = N_c T_s = (N_a + N_\eta) T_s \quad (5.98)$$

$$\tau_{SL} + \tau_{OR} = N_d T_s = (N_b + N_\eta) T_s \quad (5.99)$$

$$\tau_{OR} + \tau_{OL} = \tau_T = N_T T_s \quad (5.100)$$

where T_s denotes the sampling period, c is the wave speed, x' , x are shown in Fig. 5.11 and $x + x_o = L$.

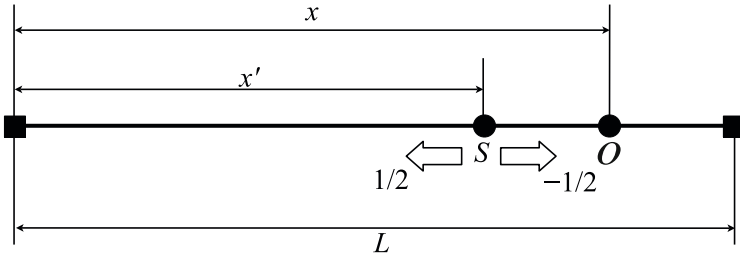


Fig. 5.11 Impulsive source excitation on finite length of string

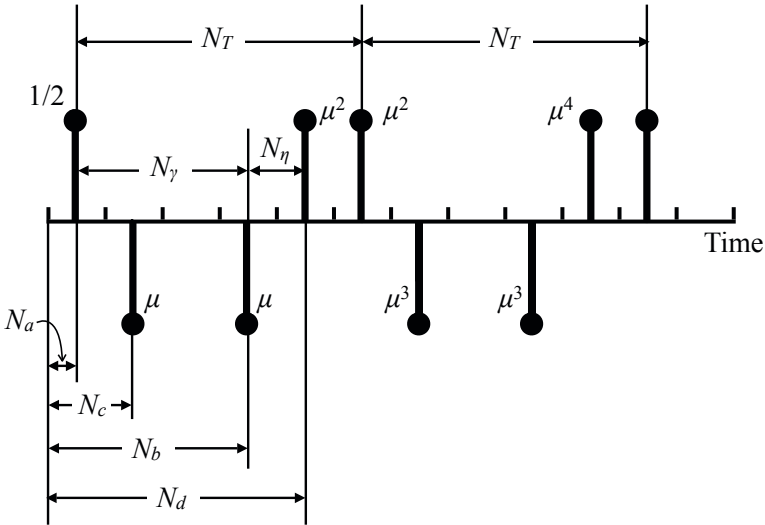


Fig. 5.12 Initial time sequence of impulse response

5.5.2 Transfer Function of Impulse Response

The impulse response can be represented by a sum of the four types of responses:

$$h(n) = h_a(n) + h_b(n) + h_c(n) + h_d(n). \quad (5.101)$$

The **transfer function**, which is defined by

$$H(z^{-1}) = \sum_n h(n)z^{-n}, \quad (5.102)$$

can be written as

$$H(z^{-1}) = \frac{1}{2} \frac{z^{-N_a} - \mu z^{-N_b} - \mu z^{-N_c} + \mu^2 z^{-N_d}}{1 - \mu^2 z^{-N_T}}, \quad (5.103)$$

where μ represents the magnitude of reflection coefficients for the boundaries.

The poles of the transfer function, which corresponds to eigenfrequencies of the string, can be determined by using the resonant characteristics of the string independent of the source and observation positions. In contrast, the zeros are determined by the locations of source or observation points independent of the resonance properties of the vibration of the string. By substituting $z = e^{i\Omega}$ for z where $\Omega = \omega T_S$ in the equation above, the **frequency response** of the string can be derived as described in the following section.

5.5.3 Frequency Response of String Vibration

Recall that μ represents the reflection coefficient, and set $\mu = e^{-\delta}$. Namely,

$$\delta = -\ln \mu \quad (5.104)$$

into Eq. 5.103, the numerator of the equation can be rewritten as

$$\begin{aligned} & (z^{-N_a} - \mu z^{-N_b} - \mu z^{-N_c} + \mu^2 z^{-N_d})|_{z=e^{i\Omega}} \\ &= e^{-i\frac{\Omega}{2}(N_a+N_b+N_\eta)} e^{-\delta} \left(e^{i\frac{\Omega_1}{2}} - e^{-i\frac{\Omega_1}{2}} \right) \left(e^{i\frac{\Omega_2}{2}} - e^{-i\frac{\Omega_2}{2}} \right), \end{aligned} \quad (5.105)$$

where

$$\Omega_1 = \Omega(N_b - N_a) - i\delta \quad (5.106)$$

$$\Omega_2 = \Omega N_\eta - i\delta. \quad (5.107)$$

Similarly, the resonant part becomes

$$\frac{1}{1 - \mu^2 z^{-N_T}} \Big|_{z=e^{i\Omega}} = \frac{e^{\delta} e^{i\frac{N_T \Omega}{2}}}{e^{i\frac{\Omega_3}{2}} - e^{-i\frac{\Omega_3}{2}}} \quad (5.108)$$

where

$$\Omega_3 = N_T \Omega - i2\delta. \quad (5.109)$$

Thus the frequency response is obtained such that

$$H(e^{-i\Omega}) = i \frac{\sin\left(\frac{N_b - N_a}{2} \Omega - \frac{i\delta}{2}\right) \sin\left(\frac{N_\eta}{2} \Omega - \frac{i\delta}{2}\right)}{\sin\left(\frac{N_T}{2} \Omega - i\delta\right)} \quad (5.110)$$

where $N_a + N_b + N_\eta - N_T = 0$.

By substituting $\Omega = \omega T_s$ and $\omega = ck$ into the equation above, the frequency response can be described as

$$\begin{aligned} H(x', x, k) &= i \frac{\left(\sin kx' - \frac{i\delta}{2}\right) \sin\left(\sin k(x - x') - \frac{i\delta}{2}\right)}{\sin(kL - i\delta)} \\ &\cong - \frac{\sin kx' \sin k(x - x')}{\delta \cos kL + i \sin kL} \end{aligned} \quad (5.111)$$

for the rigid boundary condition where x' and x are shown in Fig. 5.11. It can be found that the magnitude of the response might take its maxima at around the frequencies of free oscillation. However, such resonant characteristics are lost as the reflection coefficient becomes small.

On the other hand, the magnitude of the response becomes 0 if $\sin kx' = 0$ or $\sin kx = 0$. This indicates that the frequencies of waves cannot be excited by the positions where $\sin kx' = 0$, or the waves that become nodes at the observation point cannot be detected, even if such waves are excited on the string.

Note here also that the initial delay for the direct wave between the source and receiving positions does not necessarily have to be explicitly represented in the frequency response equation. The phase characteristics due to such a delay can be represented by the **propagation phase**[23][24] for the sinusoidal waves travelling between the source and receiving positions. These are typical characteristics for the waves progressing in a one-dimensional vibrating system.

5.5.4 Spectral Envelope for Frequency Characteristics

The numerator of the transfer function in Eq. 5.103 can be understood as the **spectral envelope** for frequency characteristics. Here the spectral envelope can also be interpreted as spectral properties of the source signal that cyclically propagates on the string. From a physical view point, the direct wave without any reflection waves is the first pulse-like wave. However, the periodical properties of the waves are determined by the group of four pulses that periodically travels on the string. Therefore, the group of pulses creates a **spectral envelope** of the signal that is "sampled" by every fundamental frequency in the spectral domain.

Note here that such spectral properties of the original wave depends on the observation position. Therefore, if the observation position is set to be very close to the source position, the spectral properties can be representative of the vibrating string under the condition of the excitation source.

5.5.5 Energy Decay Curve and Impulse Response

Suppose that an impulse response of a vibrating system is given by $h(n)$ that is assumed to be 0 when n is negative. Taking a sum of squared quantity such that

$$E_{Rev}(n) = \sum_{m=n}^{\infty} h^2(m) = \sum_{m=0}^{\infty} h^2(m) - \sum_{m=0}^{n-1} h^2(m) \quad (5.112)$$

is called a **reverberation energy decay curve** representing the reverberation decay of the system vibration energy. For example, set the impulse response as

$$h(n) = \mu^{2n} = (1 - \alpha)^n \quad (5.113)$$

following the example of impulse responses in a one-dimensional system shown in the previous section where $\mu^2 = 1 - \alpha = e^{-2\delta}$.

The energy decay curve can be written as

$$E_{Rev}(n) = \frac{(1 - \alpha)^{2n}}{1 - (1 - \alpha)^2}. \quad (5.114)$$

By introducing

$$(1 - \alpha)^{2n} = \mu^{4n} = e^{-4\delta n} \quad (5.115)$$

$$\ln(1 - \alpha) = -2\delta = -\hat{\alpha} \quad (5.116)$$

the decay curve becomes

$$E_{Rev}(n) = \frac{e^{-2\hat{\alpha}n}}{1 - (1 - \alpha)^2} = E(0)e^{-2\hat{\alpha}n}, \quad (5.117)$$

where $E(0)$ denotes the energy before starting the decay, namely, the **energy at the steady state**. It can be seen that the energy decay follows an exponential function of time. The **reverberation time** is defined as the time required for the energy to decay to 60 dB below the steady state level. Figure 5.13 shows a numerical example of the energy decay curve for a one-dimensional vibrating system. The energy decay follows an exponential function as the decay progresses.

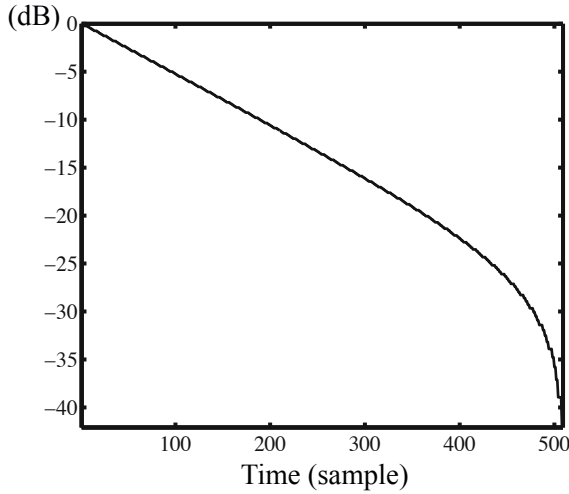


Fig. 5.13 Example of reverberation decay curve for a one-dimensional vibrating system

5.6 Eigenfrequencies and Eigenmodes

Free vibration of a finite length of string can be interpreted in two ways: based on vibration patterns composed of a fundamental and its harmonics, or following graphical representation of expansion of initial disturbance into an odd and periodic function. Eigenfrequencies and eigenmodes are important notions for uniting these ways. This section describes how a fundamental and its harmonics can be interpreted as eigenfrequencies. Consequently it can be confirmed that the period of free vibration is equal to the fundamental period, even if free vibration is composed of many harmonics. This outcome is significant for perception of sounds such as musical sounds.

5.6.1 Eigenfrequencies for Free Vibration of Finite Length of String

As illustrated in Fig. 5.6, vibration patterns with definite periods, namely normal modes of vibration, are expected in free vibration of a finite length of string. Such a vibration pattern is called an **eigenmode**, **eigenfunction**, or **modal function**. Its frequency is called an **eigenfrequency**, and it corresponds to the resonant frequency. These eigenfrequencies and eigenfunctions can be interpreted as eigenvalues and eigenvectors for a linear operator such as a matrix[2][3]. From the fact that vibration corresponding to the n -th modal function is expressed as

$$u_n(x, t) = \sin \frac{n\pi x}{L} \cos \frac{n\pi ct}{L} = \frac{1}{2} \left(\sin \frac{n\pi}{L}(x + ct) + \sin \frac{n\pi}{L}(x - ct) \right), \quad (5.118)$$

a modal function itself can be expressed as a superposition of two progressive waves with a definite frequency. The vibration above can be also rewritten in a general form such as

$$u_{\omega}(x, t) = \sin \frac{\omega}{c} x \cos \omega t. \quad (5.119)$$

Here, angular frequencies ω , however, must be a series of discrete numbers such that

$$\omega = \omega_n = 2\pi \frac{nc}{2L} \quad (\text{rad/s}) \quad (5.120)$$

where $n = 1, 2, 3, \dots$, because the function representing vibration must be periodic with an interval of $x = 2L$.

The eigenfrequencies are countless but located at every interval of the fundamental frequency. Such a series of frequencies with equal intervals is also called **harmonic structure** in terms of acoustics. The n -th angular eigenfrequency ω_n can be rewritten as

$$\omega_n = n\pi \sqrt{\frac{F_T/L}{\rho_0 L}} = \pi \sqrt{\frac{nF_T/L}{\rho_0 L/n}}, \quad (\text{rad/s}) \quad (5.121)$$

where ρ_0 (kg/m) denotes the density of a string and F_T (N) is tension of the string of interest. If we regard nF_T/L and $\rho_0 L/n$ as the tension density and partial mass of a “divided” string, the n -th eigenfrequency looks like the fundamental for the divided part.

5.6.2 Superposition of Eigenmodes and Its Period

Superposition of sinusoidal functions with harmonics is significant from an acoustical point of view. Recall that

$$\sin \omega_1 t = \sin \omega_1 (t + lT_1) \quad (5.122)$$

$$\sin n\omega_1 t = \sin n\omega_1 (t + lT_1) \quad (5.123)$$

$$T_1 = \frac{2\pi}{\omega_1} \quad (\text{s}) \quad (5.124)$$

where l is an integer. As shown in Fig. 5.14, superposition

$$u(t) = A_1 \sin(\omega_1 t + \phi_1) + A_2 \sin(2\omega_1 t + \phi_2) + \dots + A_n \sin(n\omega_1 t + \phi_n) \quad (5.125)$$

takes the same value at every T_1 seconds; namely, the signal $u(t)$ is still periodic with the fundamental period. This explains why a listener can perceive the pitch of a compound sound. In other words, the period of a signal composed of harmonic components is equal to the fundamental period regardless of whether the fundamental component is contained or not.

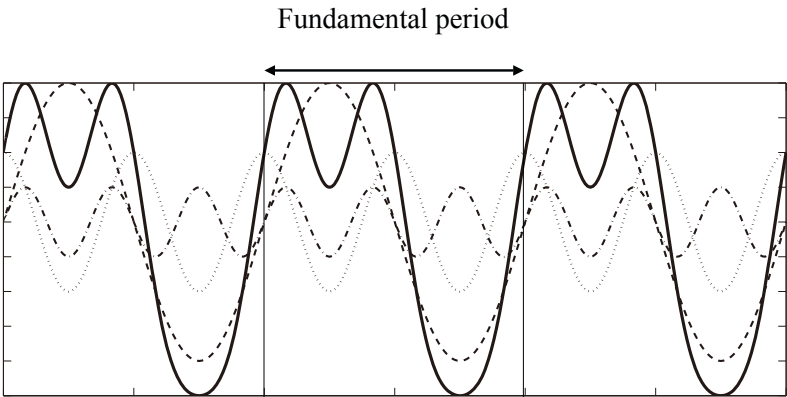


Fig. 5.14 Example of superposition of harmonic functions

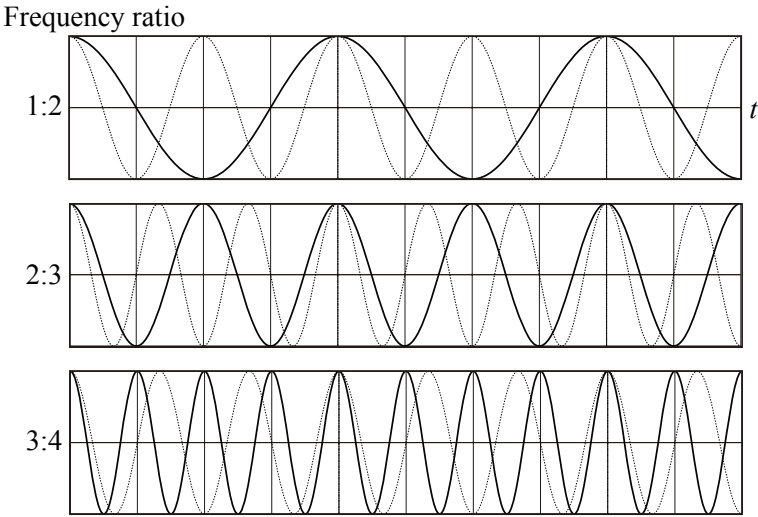


Fig. 5.15 Examples for pairs of two harmonic components from [9] (Fig.5.5)

Figure 5.15 illustrates samples of harmonic pairs of waveforms. The top panel shows waveforms of a fundamental and second harmonics, respectively. The two components return to their initial positions every fundamental period; however, the second harmonic component takes two cycles every fundamental period, while the fundamental one takes only one. Similarly, the middle one illustrates the second and third harmonics. Again both components return to the initial points every fundamental period, but two and three cycles are required for the second and third ones, respectively. The third and fourth harmonics are similarly drawn in the bottom row,

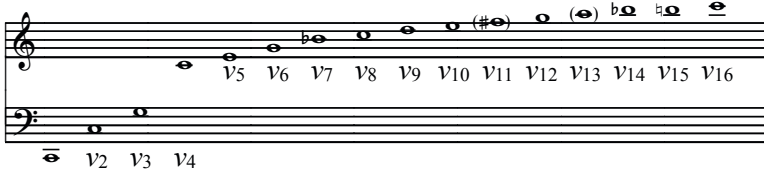


Fig. 5.16 Musical notes following fundamental and its harmonics from [4] (Fig. 2.7)

where the third one needs three cycles, and four cycles are necessary for the fourth one for every fundamental period. These samples indicate that the number of cycles contained in the fundamental period increases with the order of harmonics, and thus a pair composed of higher-order harmonics includes many cycles every fundamental period.

Consequently, a pair of lower-ordered harmonics could be consonant rather than a higher-ordered pair. Figure 5.16 shows musical notes corresponding to a fundamental and its harmonics where the fundamental is set as C[4]. Musical intervals between an adjacent harmonic pair becomes narrow as the order of harmonics increases, and thus tones become dissonant as the order becomes higher.

5.6.3 Expression of Free Vibration under Initial Disturbance

A formal expression can be introduced for free vibration of a string using superposition of modal functions. This formulation provides a physical basis for representation of a function as a periodic expression by using Fourier series in terms of mathematics. An approach to vibration analysis using Fourier series expansion, namely superposition of modal functions, is helpful for analyzing tonal characteristics of sound, which represents for example the differences between a piano and harpsichord. Fourier series expansion of a function can be geometrically interpreted as decomposition of a vector into its orthogonal basis in the space of interest[3]. Fourier coefficients correspond to the coordinates with respect to the basis. Reference [8] might be informative for the theoretical background of mathematical expression of vibration using Fourier series.

Suppose that initial conditions are set to be

$$u(x, 0) = a(x) \quad (0 \leq x \leq L) \quad (m) \quad (5.126)$$

$$v(x, 0) = 0. \quad (5.127)$$

Free vibration of a finite string can be expressed as

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x \cos(\omega_n t) \quad (0 \leq x \leq L) \quad (m) \quad (5.128)$$

under the boundary conditions such that

$$u(0, t) = 0, \quad u(L, t) = 0, \quad (\text{m}) \quad (5.129)$$

where A_n are called modal magnitudes and determined by the initial condition of $a(x)$.

The expression above indicates that the initial disturbance can be expressed as superposition of modal functions (or modal expansion of a periodically extended function) like

$$a(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x. \quad (5.130)$$

Following the expression above if the initial disturbance looks like the n -th harmonic function, then vibration could also be made of the n -th mode only. On the other hand, if the n -th harmonic function were not necessary to represent initial disturbance, then vibration might be composed of all the modal functions except the n -th mode. The modal magnitude can be expressed as

$$A_n = \frac{2}{L} \int_0^L a(x) \sin \frac{n\pi}{L} x dx, \quad (5.131)$$

which is called **Fourier coefficients**. Representation of a function as a virtually periodic function by modal expansion is called **Fourier series representation** of a function.

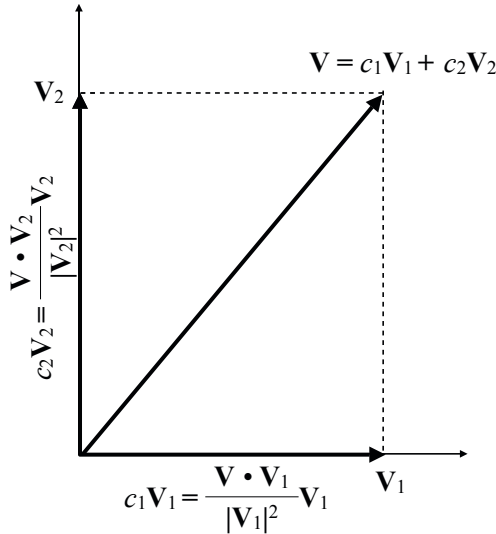


Fig. 5.17 Decomposition of vector into orthogonal vectors

Fourier coefficients, which look like the average of an integrand, normalized by the factor of $L/2$, can be understood as coordinates with respect to the basis of a space of interest, if sinusoidal functions were regarded as the basis[3]. Suppose that two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular (or orthogonal) and provide the basis of a 2-dimensional space. As shown in Fig. 5.17, vector $\mathbf{v}$ in the space can be written as

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \quad (5.132)$$

where

$$c_1 = \frac{\mathbf{v} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}, \quad (5.133)$$

$$c_2 = \frac{\mathbf{v} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}, \quad (5.134)$$

and the "dot product" of two vectors denotes the **inner product** or the **scalar product** of vectors[2][3].

Therefore, regarding integration for the modal functions as the dot product of modal functions, the Fourier coefficients can be interpreted as decomposition of a vector into the orthogonal vectors. The inner product of identical modal functions is given by

$$\int_0^L \sin^2 \frac{n\pi x}{L} dx = \frac{L}{2}, \quad (5.135)$$

which corresponds to the normalized factor of Fourier coefficients.

Figure 5.18 shows an example of wave propagation by initial displacement in a finite length of string with clamped ends. The same type of wave propagation can be seen in Fig. 5.9. This example shows a sample of simplified initial displacement, which is an even function with respect to the center of $x = L/2$. Here, recalling that integration of an odd function over its periodic interval results in 0, the integration over $x = 0$ to $x = L$ of a product of the initial disturbance and an even-ordered modal function results in zero. This is because the integrand could be regarded as an odd function in the interval between $x = 0$ and $x = L$ if the center is shifted to $x = L/2$. Consequently, all the even-ordered harmonic components are removed from the free vibration, if the initial displacement might be a virtually even function as shown by Fig. 5.18. In contrast, if the initial displacement might be an odd function, then all the odd-ordered harmonic components are removed.

According to the outcome above, suppressing definite modes, for example the fifth harmonic component with its harmonics, is possible. Figure 5.19 is an example of initial displacement for removing the fifth mode. Although it cannot be clearly seen in the figure, the fifth (including its odd-ordered higher harmonics) modal functions cannot be included in the propagating waves. This type of consideration was made in the history of musical instruments for removing dissonant harmonics. For that purpose, a position at which to give initial excitation is selected in sound instruments.

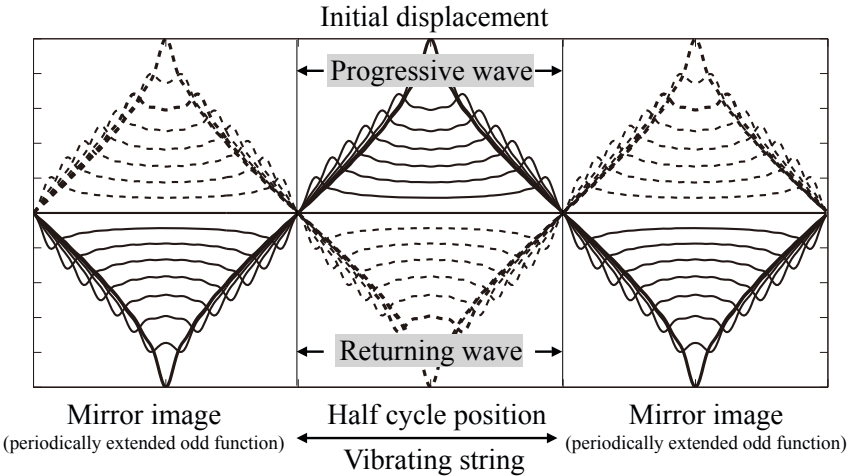


Fig. 5.18 Propagation of initial displacement obtained by modal expansion

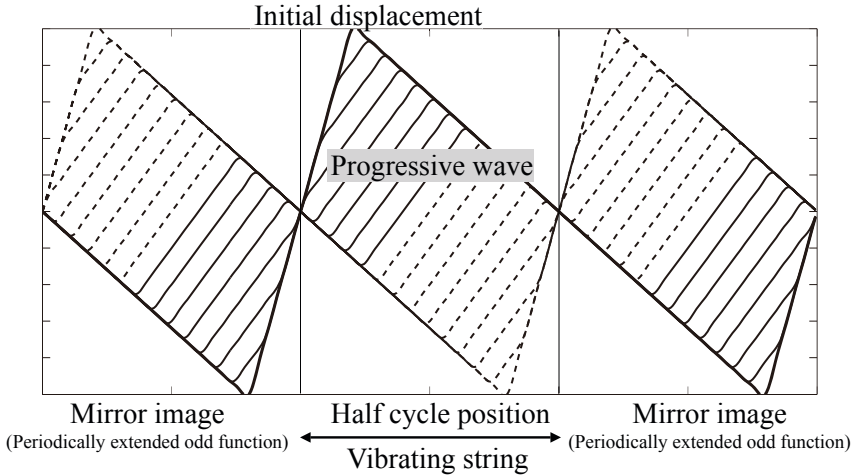


Fig. 5.19 Example of initial displacement and its propagation for removing fifth mode

As shown in Fig. 5.20, propagation of the initial velocity such as that in a piano can be also displayed following the modal expansion. Figure 5.21 is a corresponding example to Fig. 5.19 for removing the fifth modes.

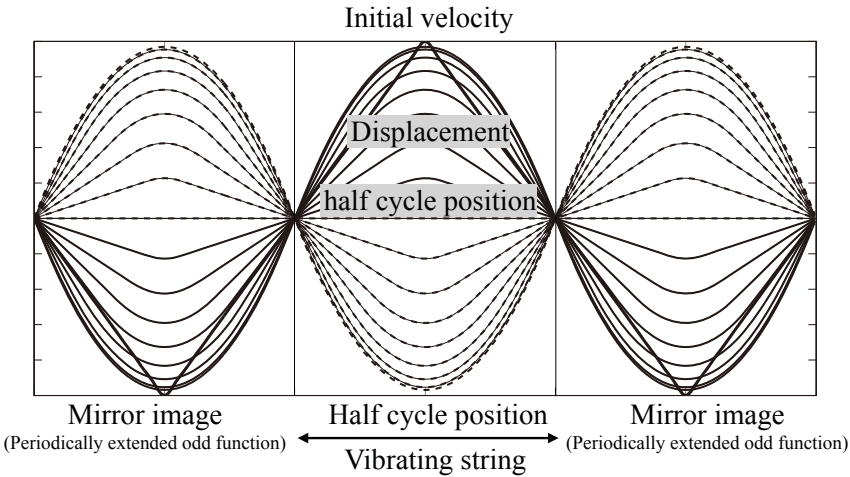


Fig. 5.20 Example of initial velocity and its propagation

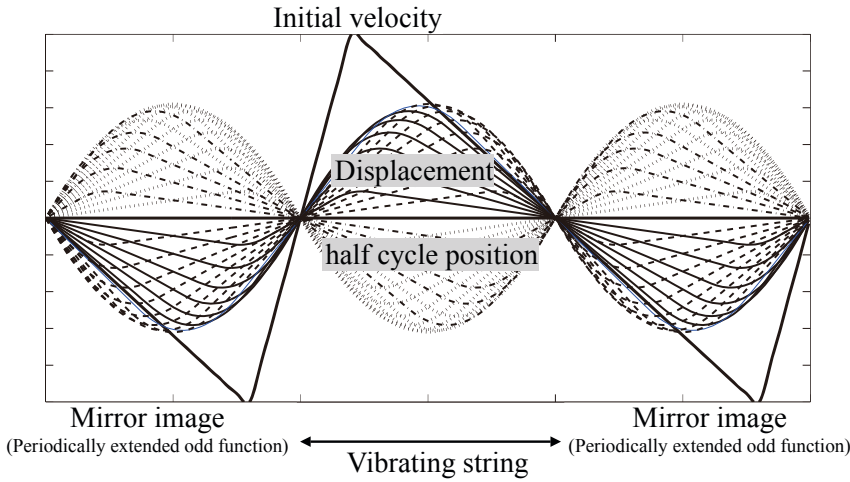


Fig. 5.21 Example of initial velocity and its propagation for removing fifth modal components